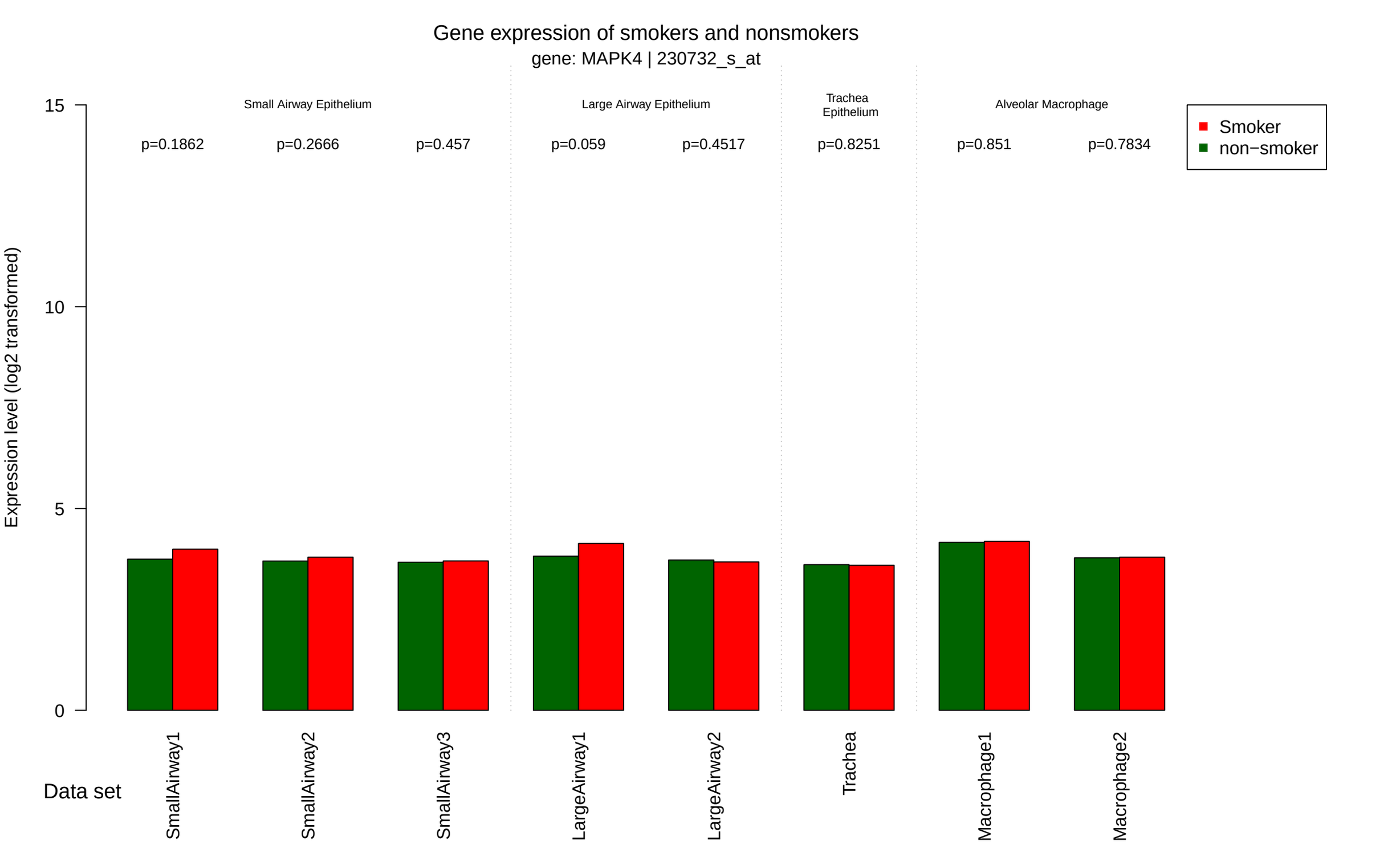
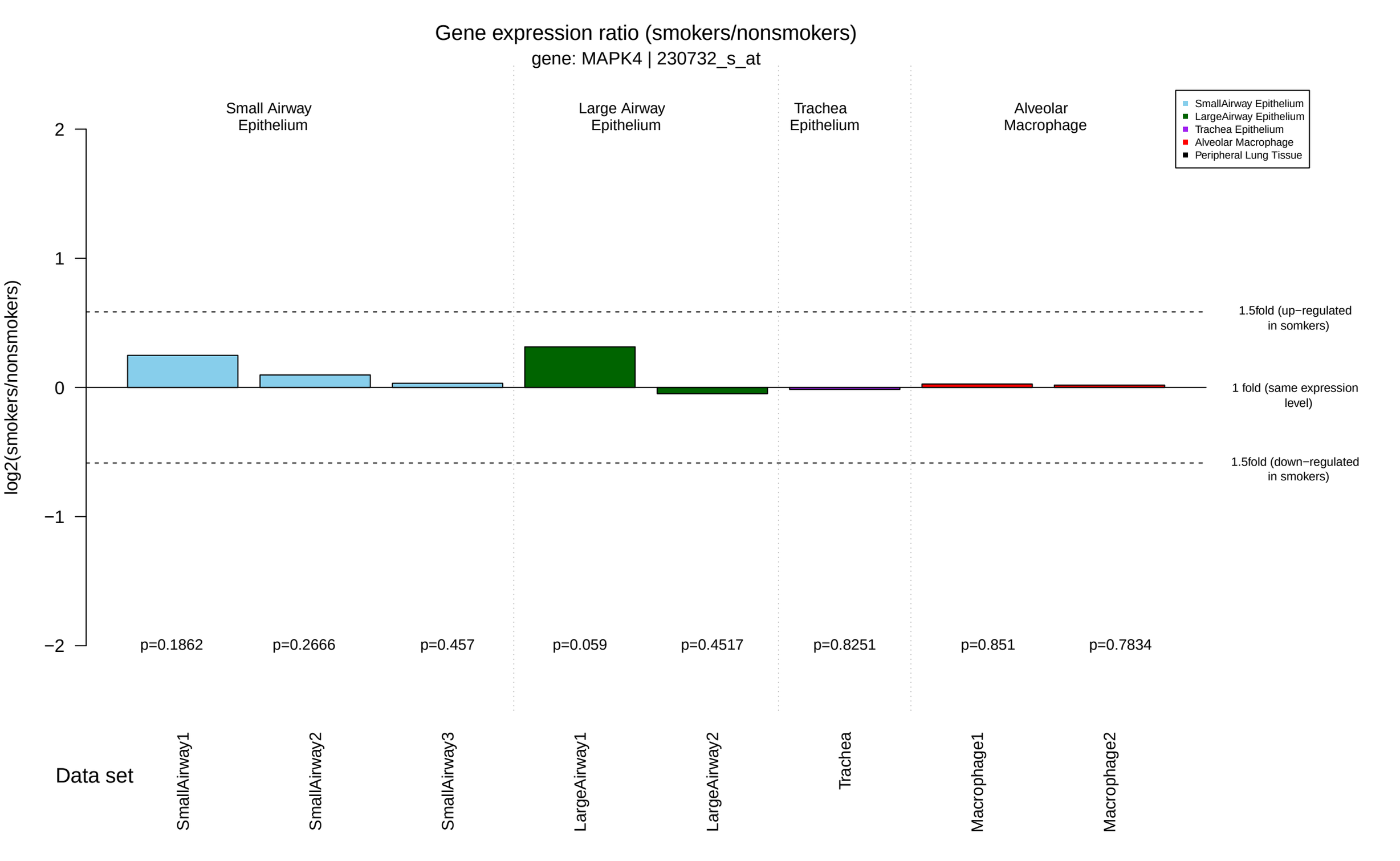
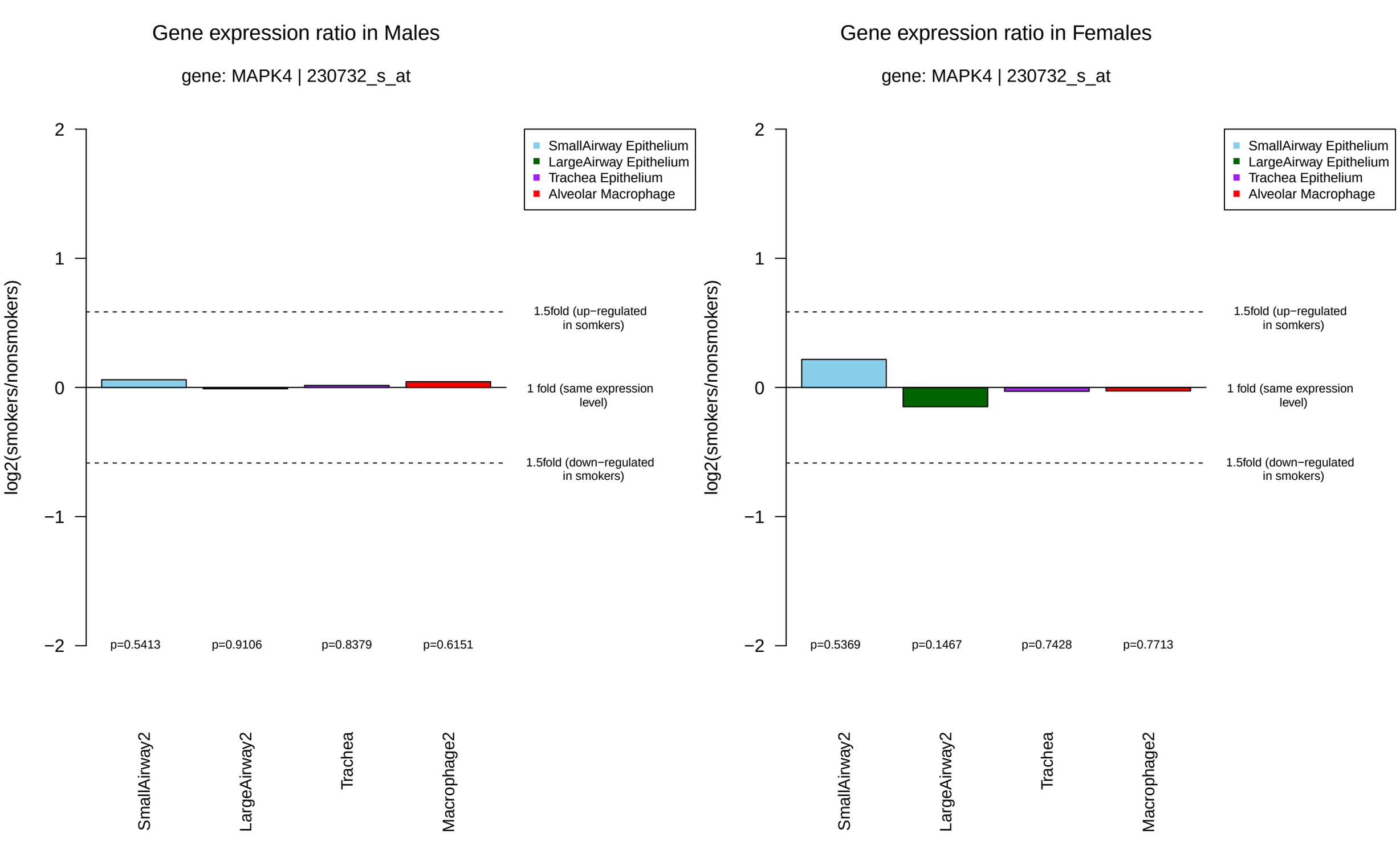
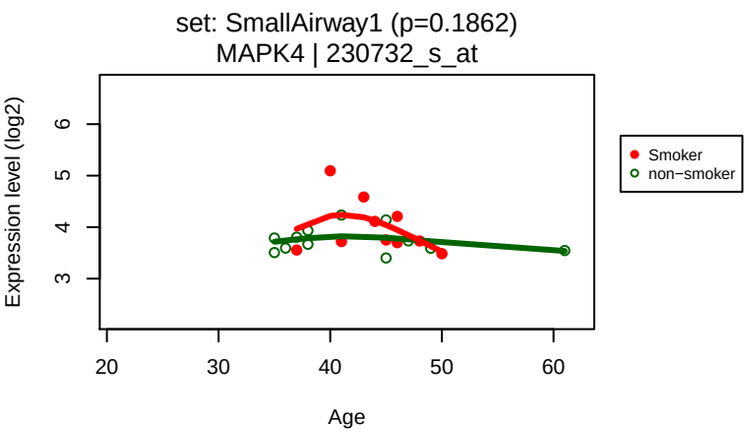
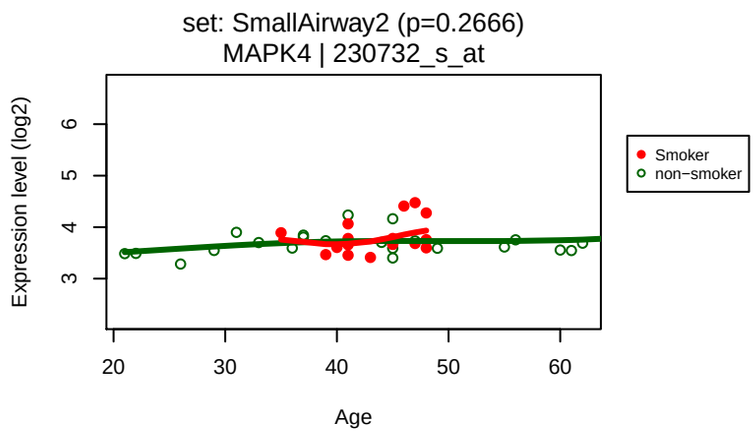
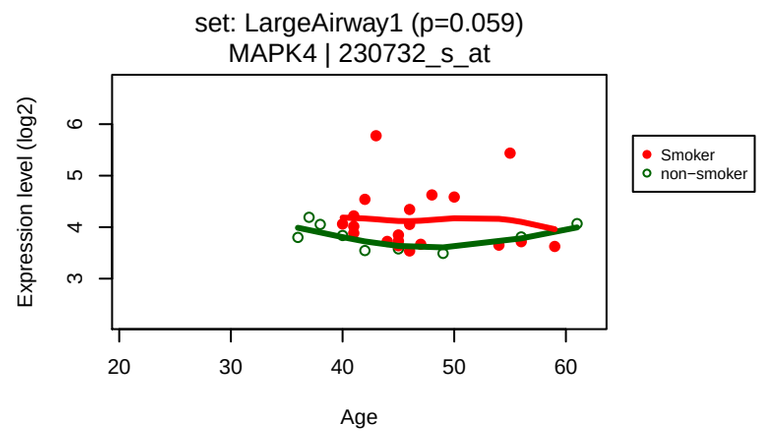

Gene expression of smokers and nonsmokers
gene: MAPK4 | 230732_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1862
p=0.2666
p=0.457
p=0.059
p=0.4517
p=0.8251
p=0.851
p=0.7834
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

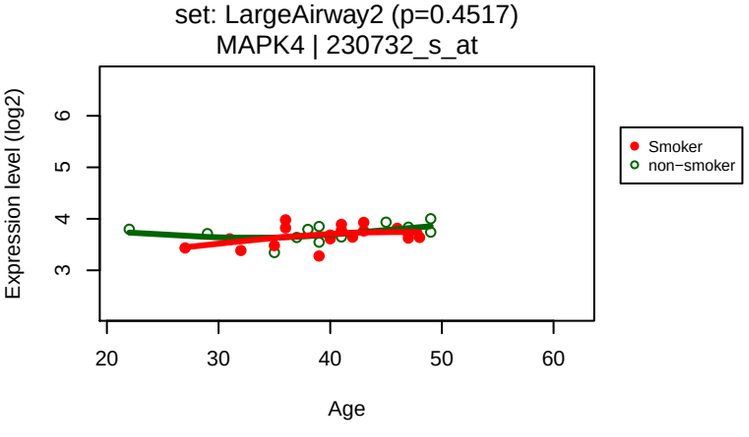

Gene expression ratio (smokers/nonsmokers)
gene: MAPK4 | 230732_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1862
p=0.2666
p=0.457
p=0.059
p=0.4517
p=0.8251
p=0.851
p=0.7834
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

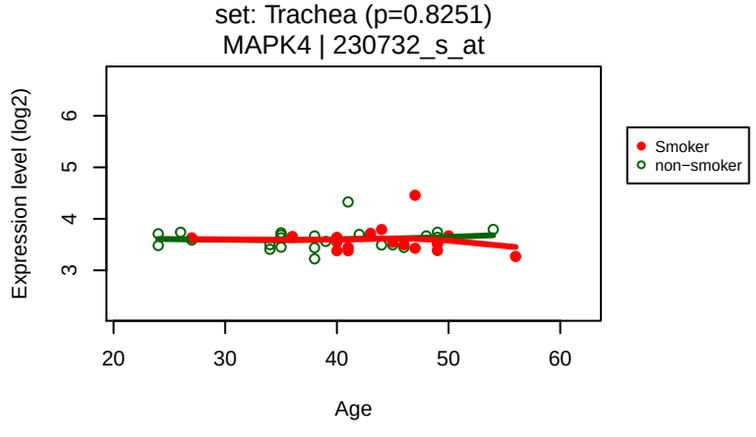

Gene expression ratio in Males
gene: MAPK4 | 230732_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5413
p=0.9106
p=0.8379
p=0.6151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK4 | 230732_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5369
p=0.1467
p=0.7428
p=0.7713
SmallAirway2
LargeAirway2
Trachea
Macrophage2

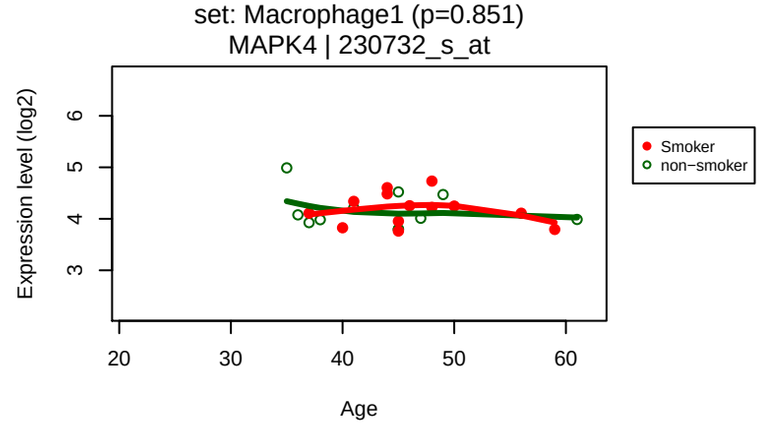

set: SmallAirway1 (p=0.1862)
MAPK4 | 230732_s_at
Expression level (log2)
Age
Smoker
non−smoker

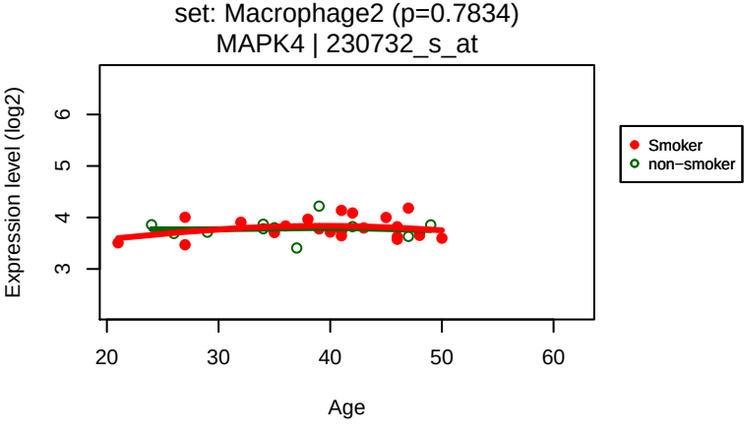

set: SmallAirway2 (p=0.2666)
MAPK4 | 230732_s_at
Expression level (log2)
Age
Smoker
non−smoker

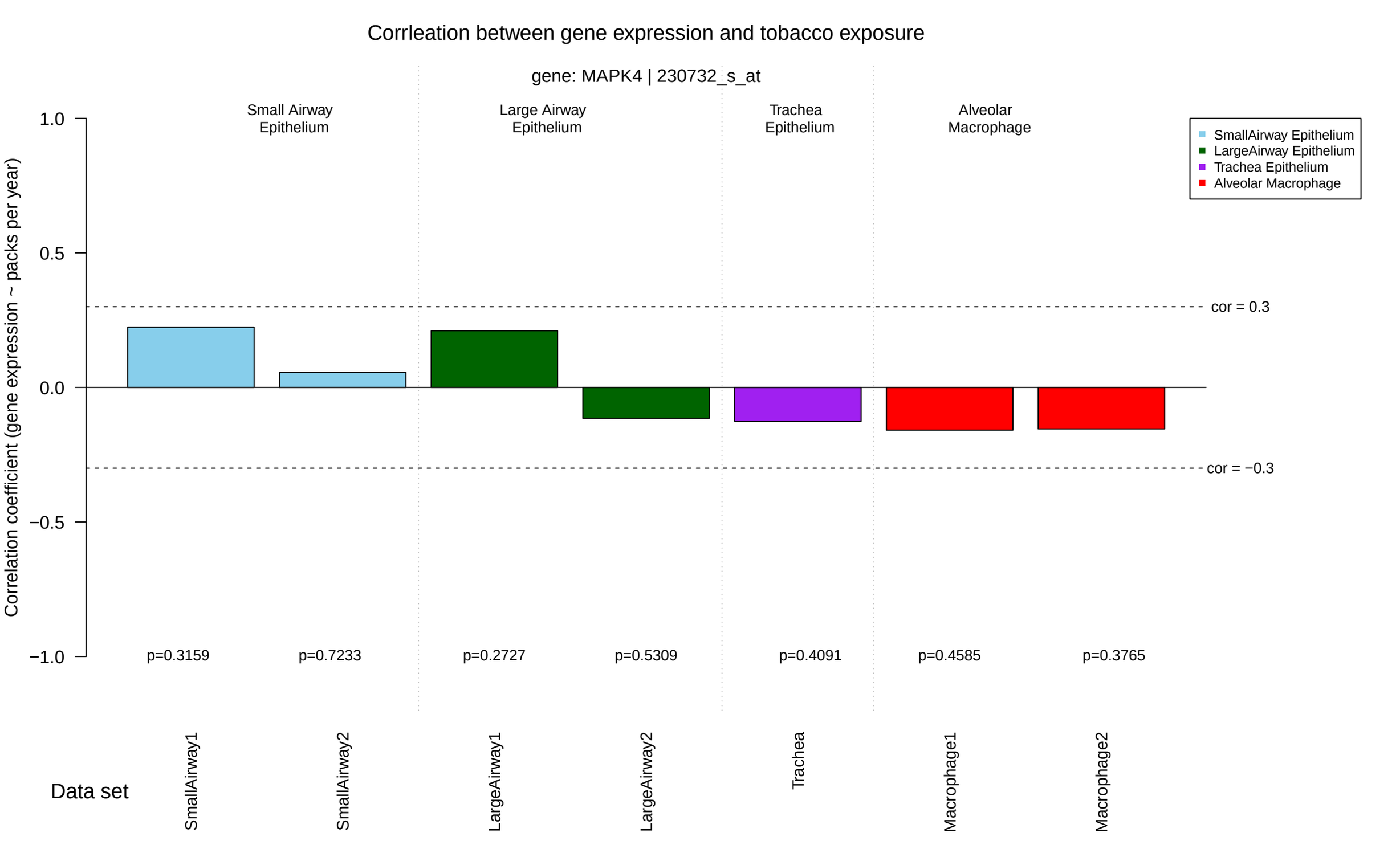

set: LargeAirway1 (p=0.059)
MAPK4 | 230732_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4517)
MAPK4 | 230732_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8251)
MAPK4 | 230732_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.851)
MAPK4 | 230732_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7834)
MAPK4 | 230732_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK4 | 230732_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3159
p=0.7233
p=0.2727
p=0.5309
p=0.4091
p=0.4585
p=0.3765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2